

AUB PayMate	Doc Name	User Operation Guide	
Technical documents	Version	V1.0	
User Operation Guide			
File name	Version	Modifier	Data

<i>User operation guide</i>	<i>Draft</i>	<i>Chris</i>	<i>20210521</i>

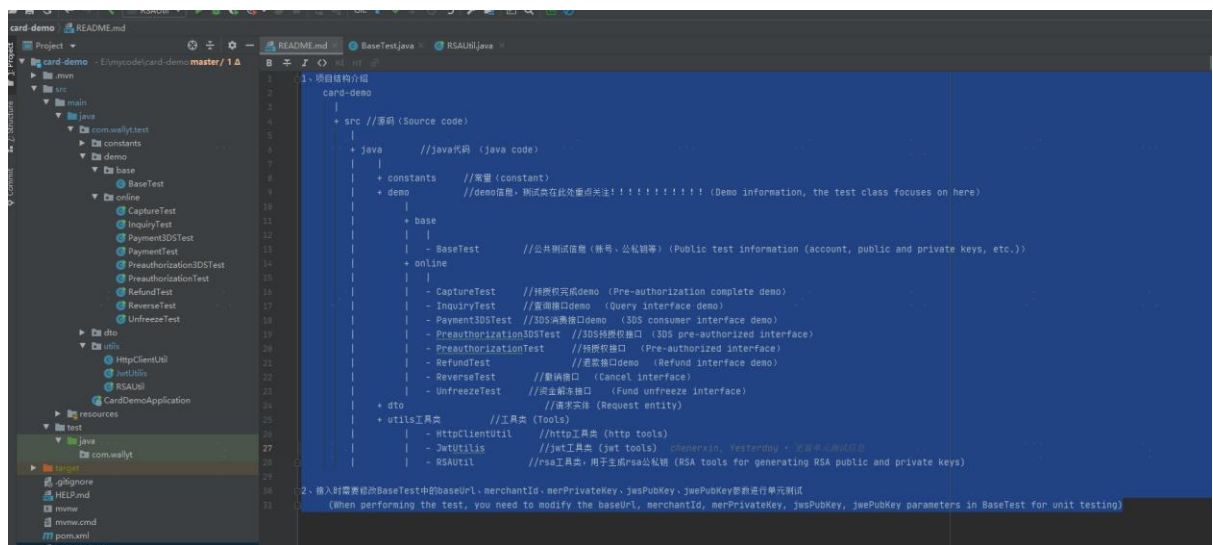
Catalog

1. DEMO FOLDER STRUCTURE.....	4
2.INTERFACE DOCUMENTATION.....	4
2.1、 READ THE DOCUMENT:	4
2.2、 To GENERATE MERCHANT PUBLIC AND PRIVATE KEY (BY USING JAVA IDEA) :	4
2.2.1 To generate merchant public and private key.....	4
2.2.2 To generate merchant public and private key.....	5
2.2.3 To generate merchant public and private key for those users that do not use java idea	5
3. IMPORT THE PROJECT AND MODIFY THE DEMO CONTENT.....	7
3.1、 IMPORT PROJECT:	7
3.2、 UPLOAD MERCHANT PUBLIC KEY: (UPLOAD MERCHANT PUBLIC KEYS IN THE MERCHANTPORTAL)	7
3.3、 DOWNLOAD INSTITUTION PUBLIC KEY:	8
3.4、 DEMO BASETest REQUEST PARAMETER VALUES MODIFICATION:	8

Introduction

For card merchants , to integrate the API more quickly, please read the following carefully.

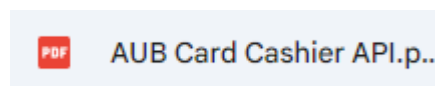
1. DEMO Folder Structure



2. Interface documentation

2.1、 Read the document:

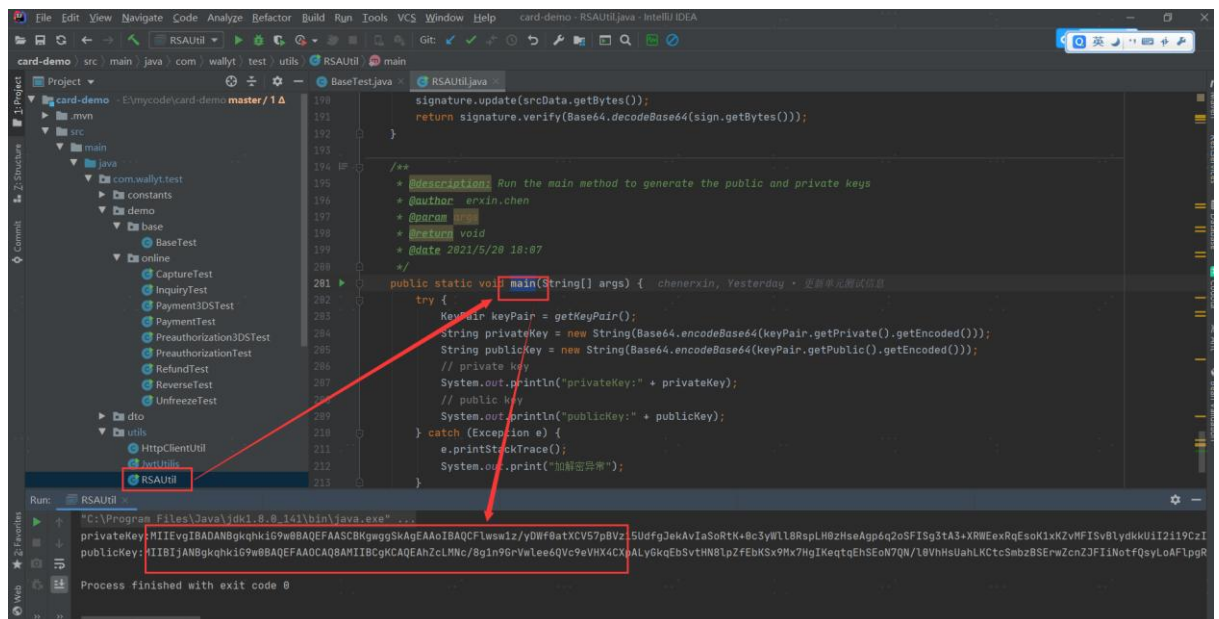
After receiving the interface document, please read the document in detail.



2.2、 Generate merchant public and private key (By using java idea):

2.2.1 Generate merchant public and private key

Run **RSASUtil** in the demo to generate a signed public key



2.2.2 Generate merchant public and private key

Repeat step 2.2.1 to generate another pair of encryption key

2.2.3 Generate merchant public and private key for those users that do not use java idea

According to the instructions of the interface document, generate public and private keys in the Linux environment (in the Linux environment, copy and paste [shift+insert key], and then enter the following three red codes until the final file is exported. end of key pair generation)

2.2.3.1 Generate rsa private key:

openssl genrsa -out rsa_private_key_2048.pem 2048

```

[root@hadoop3 ~]# openssl genrsa -out rsa_private_key_2048.pem
Generating RSA private key, 1024 bit long modulus
....+++++
.....+++++
e is 65537 (0x10001)

```

2.2.3.2 Convert rsa private key to pkcs encoding

```
openssl pkcs8 -topk8 -in rsa_private_key_2048.pem -out  
pkcs8_rsa_private_key_2048.pem -nocrypt
```

```
[root@hadoop3 ~]# openssl pkcs8 -topk8 -in rsa_private_key_2048.pem -out pkcs8_rsa_priv  
ate key_2048.pem -nocrypt
```

2.2.3.3 Generate public key

```
openssl rsa -in rsa_private_key_2048.pem -out rsa_public_key_2048.pem  
-pubout
```

```
[root@hadoop3 ~]# openssl rsa -in rsa_private_key_2048.pem -out rsa_public_key_2048.pem  
-pubout  
writing RSA key
```

2.2.3.4 Unpack

```
tar czvf key.tar.gz rsa_private_key_2048.pem rsa_public_key_2048.pem  
pkcs8_rsa_private_key_2048.pem
```

```
[root@hadoop3 ~]# tar czvf key.tar.gz rsa_private_key_2048.pem rsa_public_key_2048.pem p  
kcs8_rsa_private_key_2048.pem
```

2.2.3.5 Export to local

```
sz key.tar.gz
```

```
[root@hadoop3 ~]# sz key.tar.gz
```

2.2.3.6 After exporting, finally use the public and private keys in the red box in the figure below

名称	大小	压缩后大小
..		
pkcs8_rsa_private_key_2048.pem	916	?
rsa_private_key_2048.pem	887	?
rsa_public_key_2048.pem	272	?

2.2.3.7 Remove the blank spaces in the key(if you generate according to 2.2.3 there will be blank spaces)

```

1 -----BEGIN PRIVATE KEY-----
2 MIEVQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSJAQEAoIBAQCO7M5NEVmlqQnB
3 0ENTK6F4xbEqFXYSiRdo4zqvrqEAM/UE/asbqRcSuuIv09xHUY3hgC19wadledPL
4 dYfEWR7O/2811RcmjwNh0DOSdaFnev2usld4Tm/4q5L96yhw18GQg8yBjbHii3
5 M8ShxTWb8s0kZhs04pC7RuMcKtZ+4GtDbJ4AVgy7kCwuPzCTVZfJUjCoHp96epV
6 N/BfxpW22vIsOzcFSXAFcyCh6FgRib7b51S5ke2Ow3MLZBb7HIV786NuC4iyFqmQ
7 lq1vo5c1ARuPQLWlyweme0GDfmlwKi3kbg4U/08/cKe1Vj14KyIw0VinMHA2yn4
8 PaUA61WjAgMBAAE=CgEAAQn9+6qGvRMjr4HUCM1VHbsvebk/y7f117KLW4P1tXt
9 BK7Lhz09MDxqOjQPaU1zQoBccx6X3aX2refFwP1bkmj2uoIdLxioS7azIJZ7vynt
10 IivtnDVFyWgIEHxMIXGgPpJazAzFdgCCDviHK66gtHQ1yyRpy9WQNgG8NFz9MSz5
11 i1Jvg/LleigqWksehMYrcKXktfBtVWkoxS6pJ1ynH8HEFIImm/1aDLKo05SgdxS
12 iNuGaygLSRO1Rp9onP11H6Sq1Fi9nfhQEnK+boKAv0nM5TbS+jugtCwighG3+OJa
13 ub874Q5JLQcqn6swfD3bons7Rx962eQvUMXi/ZTVuQKBgQDe9N18T1tHh7LRd3se
14 alpwpTdDWLgYi7w5FwsbyrAar/Axa3lRhKDLfe3ka98BI54hnw5e71ILW3sA8WA0
15 flpF0N+0Rs+i8Fmcz9jPj2YzDt5IkJg6WaQESj163vV8sxMLCZNXab2jzCEkVv0x
16 NTPc4H6zfTorrNiVWD48UYoIpQKBgQDPvT/tXmRvvTvU3y/BpIeZ+99o01zGxC/4
17 5mSmrJ1qFyupmPCB0Z2Lcm9mDkA9SpRaywT1EUKHA0fD8piQC5+kqEe0eEu8t
18 coqh/KSE1ZwCvgDFS/3tno0T7CNqN9S3YqnZ9xHSpdRK5IUUdq4EfdUuFnWtY8g
19 b0PopeBkpwKBgFeLip5p9nmsvMgJCRMNEjIxm+AcM5kuWDw8vc0TsEXGc7hn5Y
20 q15civ1G0eB7oMqozpa+N6QALJpxLIpQFqJkvJvntf3PjBBDmgkfcEyaCPPLosqm
21 f1ZPTyysQeotOp/jwgir+DR9S10rQJ9S9Y5G+bUQ/QEQWkaRHy64Y01AoGBALEJ
22 yXTocmqj2N2606FmHn1rzu7N0C3h2MLiBdOScJBncCm9aLOJjLR0TP1fg9mJfE
23 HXUVN7X30kQJKDj+Tr4x0DxkjKLVG5ZQL9ugBAttQ6pPCLqBvnyvK2T/QLTn1jE
24 n5mB9hCe//TsGXc9RkXrtchj7T0N83NjIFkICcXVAoGAHd4v4j2qWt1400GWwRk9
25 NsVRlXmaTacsS1KneNMk6j2HnryMQbvTeQp5TOSrpDIvozxNVFIbngRdgJ+kXIP
26 r802ZL6mDMkGEB7FHSLO9DMF/TmPW0lbt3EJ/BA+4/Vsdc7czJQ4ajIy/bXDou2C
27 GLKRO8q75kvZKw0StmVede4=
28 -----END PRIVATE KEY-----
29

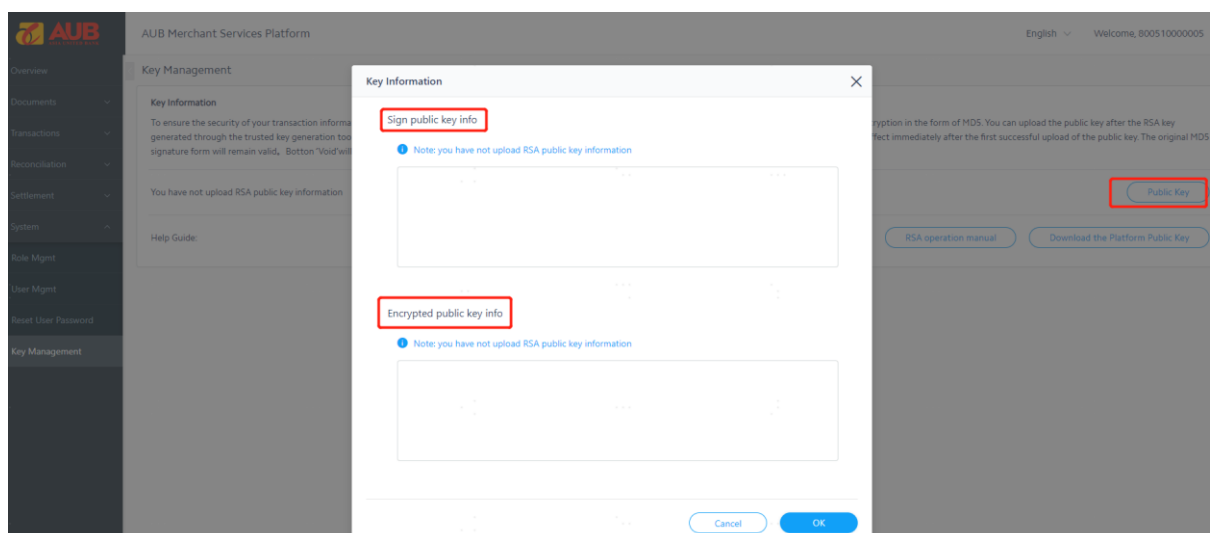
```

3. Import the project and modify the demo content

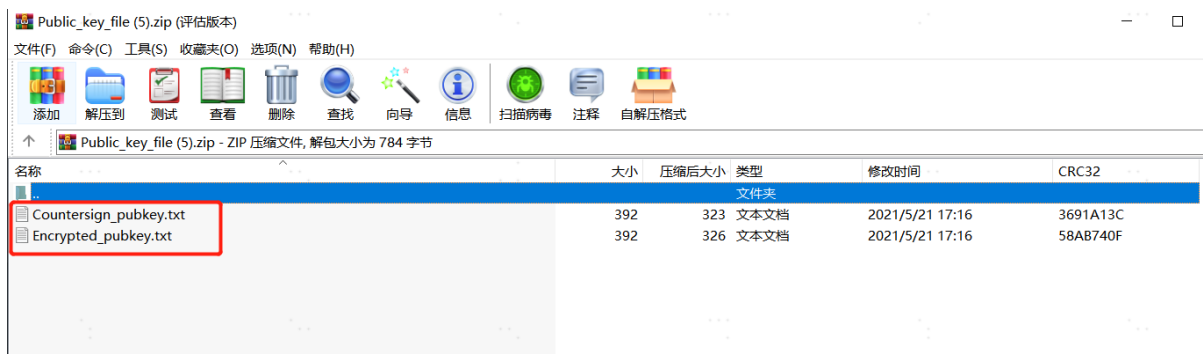
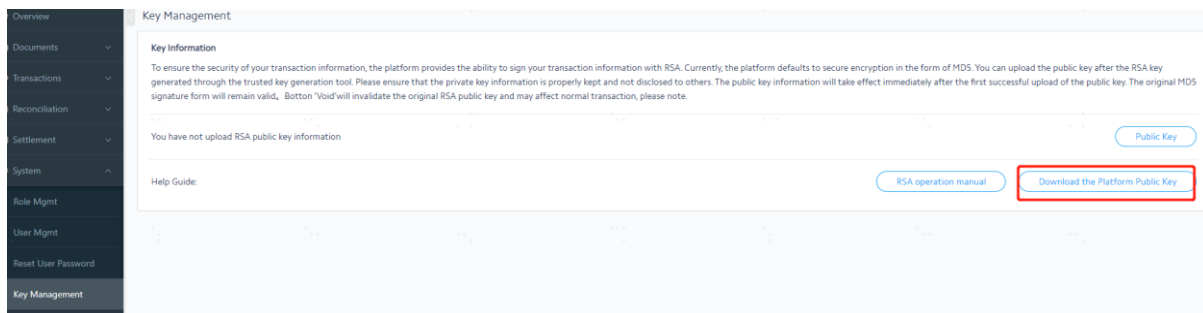
3.1、 Import project:

We provide the card-demo-v1.0 project under IDEA/eclipse, which can be used for your testing.

3.2、 Upload merchant public key: (Upload merchant public keys in the merchant portal)

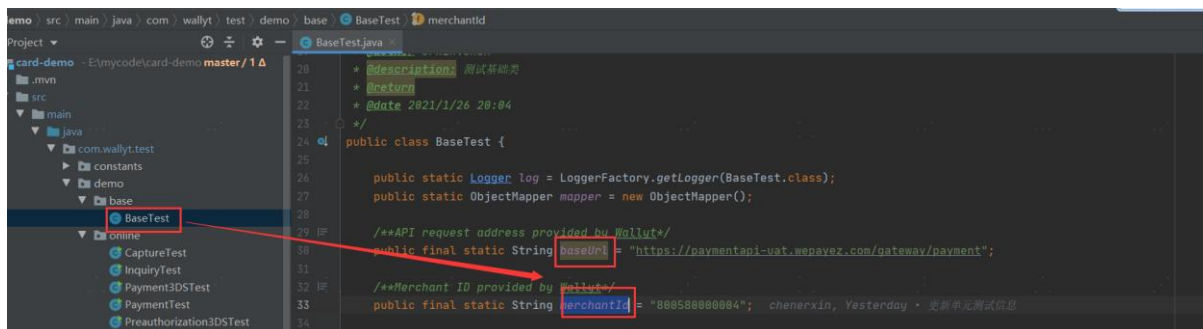


3.3、Download Institution public key:

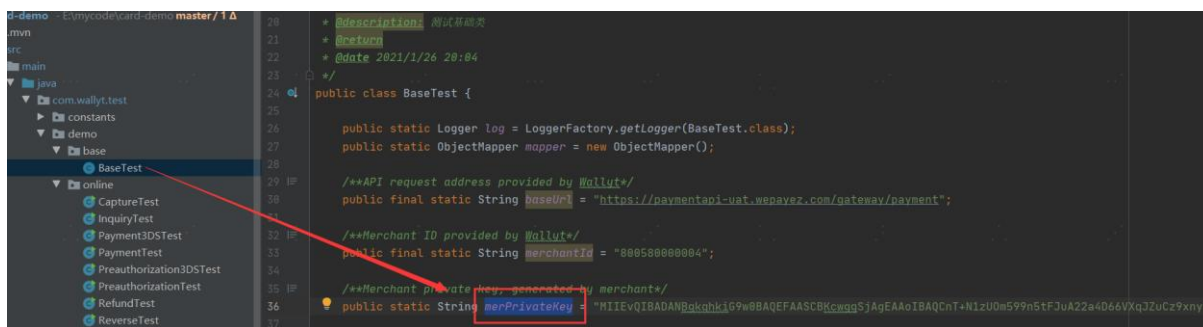


3.4、Demo BaseTest request parameter values modification:

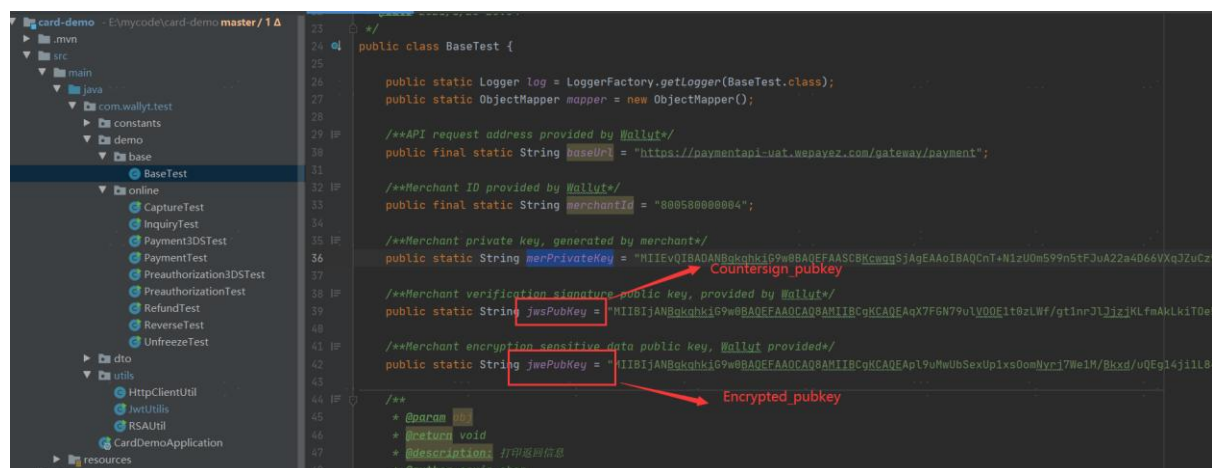
3.4.1 Replace baseUrl、merchantId:



3.4.2 Replace demo merPrivateKey with merchant private key privateKey generated in 2.2



jwePubKey corresponds to Encrypted_pubkey downloaded in 3.3



The screenshot displays an IDE interface with two main panels. On the left is the 'Project Explorer' showing a hierarchical view of the project structure. The project is named 'card-demo' and is located at 'E:\mycode\card-demo'. The structure includes 'main' and 'resources' folders. Under 'main', there are 'com.wallyt.test' and 'demo' packages. The 'demo' package contains a 'base' folder with 'BaseTest' and 'PaymentTest' classes. The 'PaymentTest' class is highlighted with a red box, and a red arrow points from this box to the 'PaymentTest' class in the Source Editor. On the right is the 'Source Editor' showing the code for 'PaymentTest.java'. The code includes imports for 'java.util.Map' and 'java.util.HashMap', and annotations for 'Author' (chenexin), 'Description' (消费接口demo), and 'Date' (2021/5/7 10:33). The class 'PaymentTest' extends 'BaseTest' and contains a 'main' method. The 'main' method calls 'PaymentTest.main()' and 'PaymentTest.main()' with coverage. The code also includes a 'PaymentCardReq' object and a 'PaymentCustomerInformationReq' object.